

**Part C**

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**Course title: Bayes Methods**  
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1. Let  $\theta \in \mathbb{R}^p$  be a  $p$ -component parameter vector with prior  $\pi(\theta)$  and  $y \in \mathbb{R}^n$  an  $n$ -component data vector with observation model  $y \sim p(y|\theta)$ . the parameters are positive, and satisfy an order constraint,  $0 < \theta_1 < \theta_2 < \dots < \theta_p < \infty$ .

(a) [13 marks]

- (i) Consider the following proposal scheme for an MCMC algorithm targeting  $\pi(\theta|y)$ . Let  $\theta_0 = 0$ . At each step choose an index  $i \sim U\{1, 2, \dots, p\}$  at random and set  $\theta'_j = \theta_j$  for each  $j \neq i$ ;
  - if  $i < p$  set  $\theta'_i \sim U(\theta_{i-1}, \theta_{i+1})$ ;
  - if  $i = p$  set  $\theta'_p = \theta_{p-1} + \delta$  with  $\delta \sim \text{Exp}(\lambda)$  and  $\lambda > 0$  a fixed MCMC parameter.
 Give an MCMC algorithm using this proposal scheme. Give the acceptance probabilities in as much detail as you can.
- (ii) Now consider the following proposal. Draw  $u_1 \sim U(1/2, 2)$  and  $u_2 \sim N(0, \sigma^2)$  where  $\sigma > 0$  is a fixed parameter of the MCMC. Set  $\theta' = u_1\theta + u_2$ , that is  $\theta'_i = u_1\theta_i + u_2$ ,  $i = 1, 2, \dots, p$ . Calculate the acceptance probability  $\alpha(\theta'|\theta)$  in as much detail as you can.
- (iii) The proposal schemes in Questions 1(a)i and 1(a)ii are combined in a single MCMC algorithm which alternates between the two updates. Comment briefly on any advantage this might give over using the scheme in Question 1(a)i or Question 1(a)ii alone.
- (iv) State briefly how you would choose values for  $\lambda$  and  $\sigma$ , and how you would check for MCMC convergence.

- (b) [6 marks] Consider a Markov chain simulated in the following way. Suppose  $\theta^{(0)} \sim \pi(\theta^{(0)})$  is a draw from the prior and for  $t = 0, 1, 2, \dots$  we generate a Markov chain by simulating  $y^{(t)} \sim p(y^{(t)}|\theta^{(t)})$  and then  $\theta^{(t+1)} \sim \pi(\theta^{(t+1)}|y^{(t)})$ .

- (i) Calculate the joint density of  $\theta^{(0)}, \theta^{(1)}$  and show that they are exchangeable.
- (ii) Give the transition probability density  $K(\theta, \theta')$  for the chain and show the chain is reversible with respect to a stationary distribution which you should give.

- (c) [6 marks] Suppose  $p = 1$  now so  $\theta$  is a scalar parameter. Suppose we simulate pairs  $(\phi_i, \theta_i)$ ,  $i = 1, 2, \dots, K$  as follows: parameter  $\phi_i \sim \pi(\phi_i)$  is an independent draw from the prior; synthetic data  $y'_i \sim p(y'_i|\phi_i)$  is an independent draw from the observation model; the Markov chain in Question 1(a)iii is initialised with a draw  $\theta_i^{(0)} \sim \pi^{(0)}$  from an initial distribution  $\pi^{(0)}$  and run for  $T$  steps, and we set  $\theta_i = \theta_i^{(T)}$  equal the last state. Let  $\phi = (\phi_1, \dots, \phi_K)$  and  $\theta = (\theta_1, \dots, \theta_K)$  be samples generated in this way.

- (i) Suppose  $P(\theta^{(T)}|\theta^{(0)}) = \pi(\theta^{(T)}|y)$  (ie, the chain has converged by  $T$  steps for all starting states  $\theta^{(0)}$ ). Let  $p(\phi, \theta)$  be the joint distribution of the random vectors  $\phi$  and  $\theta$ . Show that  $p(\phi, \theta) = p(\theta, \phi)$ .
- (ii) Briefly outline a test for MCMC convergence which makes use of the result in Question 1(c)i. State any further assumptions you make to frame your test.

2. (a) [5 marks] Let  $X_1, X_2, X_3, \dots$  be an infinite exchangeable sequence of binary random variables.

(i) What is an infinite exchangeable sequence?

(ii) Let

$$p(x_1, \dots, x_n) = \Pr(X_1 = x_1, \dots, X_n = x_n).$$

What form does de Finetti's theorem give for the joint distribution  $p(x_1, \dots, x_n)$ ?

- (b) [10 marks] A Polya urn contains  $b$  black and  $w$  white balls. At step  $i = 1, 2, 3, \dots$ , a ball is drawn uniformly at random from the urn. A variable  $X_i$  records the color,  $X_i = 1$  if the  $i$ th ball is black, and otherwise  $X_i = 0$ . The ball is returned to the urn along with 1 ball of same color.

(i) Write down the probability for the sequences 1000 and 0001.

(ii) Show that  $X_1, X_2, X_3, \dots$  is an infinite exchangeable sequence.

(iii) Let  $p(x_1, x_2, \dots, x_n) = \Pr(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$ . Show that

$$p(x_1, \dots, x_n) = \int \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \text{Beta}(\theta; b, w) d\theta.$$

*Hint: Note that  $x\Gamma(x-1) = \Gamma(x)$ , and the Beta probability density is*

$$\text{Beta}(\theta; u, v) = \frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)} \theta^{u-1} (1-\theta)^{v-1}.$$

(iv) Let  $0 < m < n$  be fixed positive integers. Calculate  $p(x_{m+1}, \dots, x_n | x_1, \dots, x_m)$ , the posterior predictive probability for the outcome  $X_{m+1} = x_{m+1}, \dots, X_n = x_n$  given data  $X_1 = x_1, \dots, X_m = x_m$ .

- (c) [10 marks] A random distribution  $G \sim DP(\alpha, H)$  on  $\Omega = [0, 1]$  has a Dirichlet Process distribution with parameter  $\alpha$  and base distribution

$$H(d\phi) = \text{Beta}(\phi; a, b) d\phi.$$

Conditional on  $G$ , the parameters  $\theta_i \sim G$ ,  $i = 1, 2, \dots, n$  are independent and identically distributed according to  $G$ . Let  $\theta = (\theta_1, \dots, \theta_n)$ . Data vector  $x = (x_1, x_2, \dots, x_n)$  has observation model  $x_i \sim \text{Bernoulli}(\theta_i)$  independently for  $i = 1, 2, \dots, n$ . For given  $\theta$ , let  $S = \{S_1, S_2, \dots, S_K\}$  be a  $K$  component partition of  $\{1, 2, \dots, n\}$  and  $\theta^* = (\theta_1^*, \dots, \theta_K^*)$  a parameter vector  $\theta^* \in [0, 1]^K$  satisfying  $\theta_i = \theta_k^*$  for each  $k = 1, 2, \dots, K$  and all  $i \in S_k$ . Assume  $\alpha$  and  $H$  are given.

(i) Give a definition of a Dirichlet Process.

(ii) Write down the joint posterior distribution  $\pi(S, \theta^* | x)$  for  $S$  and  $\theta^*$  given  $x$ .

*Hint: the Chinese Restaurant process determines a distribution  $P(S)$  over partitions  $S = \{S_1, S_2, \dots, S_K\}$  of the form  $P(S) \propto \alpha^K \prod_{k=1}^K \Gamma(n_k)$  where  $n_k = \text{card}(S_k)$  is the number of elements in  $S_k$ .*

(iii) Calculate the marginal probability  $P(S|x)$  for the partition  $S$  given the data  $x$ .

3. During the 19th century there were many fatalities in coal mines. The fatality rate might be expected to drop following the introduction of new safety laws over the course of the century but might rise when extra production was demanded during wars.

For  $i = 1, 2, \dots, n$ , let  $y_i \in (0, 100)$  be the time of the  $i$ 'th fatality. Let  $y = (y_1, \dots, y_n)$ . It is assumed that these times are events in a Poisson point process of rate  $\lambda(t)$  deaths/year which was piece-wise constant between change points. Let  $\theta_0 = 0$  and  $\theta_m = 100$  and for  $i = 1, \dots, m-1$  let  $\theta_i \in (0, 100)$  be an increasing sequence of change points at which  $\lambda(t)$  jumps up or down. The rate function is

$$\lambda(t) = \sum_{i=1}^m \lambda_i \mathbb{I}_{\theta_{i-1} < t \leq \theta_i}$$

so that  $\lambda(t) = \lambda_i$  for  $t$  in the time interval between  $\theta_{i-1}$  and  $\theta_i$ . Denote by  $\theta = (\theta_1, \dots, \theta_{m-1})$  and  $\lambda = (\lambda_1, \dots, \lambda_m)$  the vectors of unknown time and rate parameters. The number of intervals  $m$  is unknown. Suppose the change-points  $\theta_i, i = 1, \dots, m-1$ , are modeled as arrivals in a Poisson process of unknown rate  $\rho$  with units "change-point events per year".

(a) [11 marks]

- (i) Write down the prior for  $m$  and  $\theta$  given  $\rho$ , and the parameter space for  $(m, \theta, \lambda)$ .
- (ii) Data for the annual fatalities 1790-1799 and 1901-1910 are available. Mining companies were conservative and the kind of significant change in mining methods which would lead to a measurable change in fatality rates cannot have been common. Explain how you would elicit a prior for  $\lambda$  given  $m$  and a prior for  $\rho$ , stating any assumptions you make and any further information you need to complete this task.
- (iii) Write down the posterior  $\pi(\lambda, \theta, m, \rho|y)$  in terms of the available model elements. What is the parameter space for this posterior?

(b) [7 marks] Suppose you have a Metropolis Hastings MCMC algorithm targeting  $\pi(\lambda, \theta, m, \rho|y)$ , realising samples  $\lambda^{(k)}, \theta^{(k)}, m^{(k)}, \rho^{(k)}, k = 1, \dots, K$  returned from the MCMC.

- (i) Suppose the unknown true fatality rate at time  $t$  is  $\lambda^*(t)$  and the unknown true number of change points is  $m^* - 1$  (so  $\lambda^*(t)$  is piecewise constant over  $m^*$  intervals). Suppose the loss function for estimating  $\lambda^*(t)$  at  $t \in (0, 100)$  with  $\lambda(t)$  is  $(\lambda^*(t) - \lambda(t))^2$ , and the loss for estimating  $m^*$  with  $m$  is  $\mathbb{I}_{m^*=m}$ . State briefly how you would estimate  $m^*$  and  $\lambda^*(t)$  minimising the expected posterior loss for each quantity, using the MCMC output.
- (ii) How would you make a goodness of fit check for this problem? Give the details of any procedures or test statistics you would use.

(c) [7 marks] Now suppose the  $\lambda$  prior has the form  $\pi_\Lambda(\lambda) = \prod_{i=1}^m \pi(\lambda_i)$  and, in all changes to  $\lambda$  in the MCMC algorithm targeting  $\pi(\lambda, \theta, m, \rho|y)$ , the update changes a single  $\lambda_i$ -value, with the proposal  $\lambda'_i$  a draw from the prior. There is an hypothesis that  $\lambda(t)$  is decreasing, so that  $\lambda_{i+1} < \lambda_i$  for  $i = 1, \dots, m$ . Let  $\Lambda'_m = \{\lambda \in [0, \infty)^m : \lambda_{i+1} < \lambda_i, i = 0, \dots, m\}$  and  $\Lambda = \bigcup_{m=1}^{\infty} \Lambda_m$ .

- (i) How would you modify the MCMC algorithm to target  $\pi(\lambda, \theta, m, \rho|y, \lambda \in \Lambda')$ ? It is not necessary to give details of revised acceptance probabilities.
- (ii) Denote by  $B_{1,2}$  the Bayes factor measuring the support for model (1)  $\lambda \in \Lambda'$  over model (2)  $\lambda \in \Lambda$ . Show that

$$B_{1,2} = \frac{P(\Lambda'|y)}{\mathbb{E}(1/m!)}$$

where  $P(\Lambda'|y)$  is the probability  $\lambda \in \Lambda'$  in the constrained model and  $\mathbb{E}(1/m!)$  is an expectation which you should define.